Conformal Replay for Off-Policy Multi-Agent Reinforcement Learning

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ABSTRACT

In off-policy multi-agent reinforcement learning (MARL), each agent perceives the other learners as part of its environment; as their policies drift, the perceived dynamics become non-stationary and previously collected transitions turn misleading. The usual remedy—shrinking the replay buffer—discards the sample efficiency that motivates off-policy learning. We recast the *staleness* of a stored transition as its *conformity* to the other agents' *current* policies and study it with conformal prediction: a distribution-free non-conformity score, calibrated on fresh transitions into p-values, measuring how surprising a transition is under the present (or, when policies are unavailable, under a model of the currently perceived dynamics). Three findings follow. *Theory*: the value off-policy MARL targets is a perceived best-response value, and reweighting replay towards currently-compatible transitions computes the best response of a *support-starved* perceived MDP—so naively *filtering* for compatibility can move the fixed point away from the optimum by starving exploration, which we confirm empirically. *Certified replay*: where the importance ratio is available, ρ-weighted conformal and time-decay calibration *correct* the replay distribution with finite-sample guarantees instead of filtering it. *Detection* (our most robust contribution): a conformal test martingale flags when the perceived environment has drifted, with an *anytime* false-alarm guarantee via Ville's inequality, and drives a two-channel controller whose frozen alarm preserves the guarantee while an adaptive channel reads the drift rate; on standard cooperative benchmarks it detects co-learner drift faster than a TD-error heuristic at matched false-alarm, and the same martingale *certifies* the replay correction. We study the approach on controlled environments built to isolate perceived non-stationarity.

CCS CONCEPTS

• **Computing methodologies** → **Multi-agent reinforcement learning**; **Sequential decision making**; • **Mathematics of computing** → *Hypothesis testing and confidence interval computation.*

KEYWORDS

Multi-agent reinforcement learning; Off-policy learning; Experience replay; Non-stationarity; Conformal prediction; Sequential change detection

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1 INTRODUCTION

Many domains—robotics, logistics, healthcare, finance, and strategy games—could benefit from agents that learn to act together. Yet, in spite of impressive results on benchmarks such as the StarCraft Multi-Agent Challenge [12], multi-agent reinforcement learning (MARL) remains hard to deploy because of slow and unstable convergence [7].

A large part of this difficulty has a single, well-identified cause. Single-agent reinforcement learning assumes a *stationary* environment. In MARL, from the point of view of one agent, the other agents are part of the environment; as they update their policies, the dynamics that the agent perceives *change over time*. This *perceived non-stationarity* [7] is particularly damaging for *off-policy*, value-based methods, which owe their sample efficiency to a *replay buffer*: they repeatedly learn from transitions collected long ago, under *outdated* policies of the other agents. A common workaround is to keep the buffer small, so that stale transitions are quickly forgotten—but this throws away exactly the sample efficiency and the offline-learning capability that motivate off-policy learning.

The key observation of this paper is that *old does not imply useless*. A stored transition is harmful only insofar as the joint action taken by the other agents in it is *implausible under their current policies*. If their stored behaviour is still compatible with what they would do now, the transition is as informative as a fresh one. The problem of reusing the past therefore becomes one of *measuring each transition's conformity to the present*—and, as we will show, using that signal to *detect* when the buffer has drifted and to *correct* the replay distribution, rather than merely to *filter* it, which can backfire by starving exploration.

Prior work makes this concrete through an importance-sampling ratio between the current and past probabilities of the other agents' actions [3]. We revisit that idea, fix its mathematical formulation, and show that the ratio is often either *unavailable* (partial observability, continuous actions, parametric exploration, black-box teammates) or *numerically unstable* (its variance explodes). We then replace the explicit ratio by a *calibrated, distribution-free measure of conformity* drawn from conformal prediction [1, 21].

Contributions.

(1) **Theory.** We formalise the value off-policy MARL implicitly targets as a *perceived best-response value* under a quasi-stationary approximation (Section 3, Appendix B), and prove

a *support-starvation* result: reweighting replay towards currently-compatible transitions computes the best response of a support-*starved* perceived MDP, matching the true best response only for exact-ratio soft weights with full support—so naive compatibility *filtering* can move the fixed point away from the optimum (Theorem 4.2).

(2) **Certified replay.** We give a corrected account of importance-ratio reweighting (change-of-measure and its support condition, the heavy-tailed instability of direct loss correction, and the self-normalised nature of ratio prioritisation), and turn it into *calibrated* corrections with finite-sample validity—ρ-weighted conformal p-values and time-decay calibration—that *correct* the replay distribution rather than merely filter it (Sections 3, 4.5).

(3) **Detection (headline).** We introduce a conformal *test martingale* that detects drift of the perceived environment with an *anytime* false-alarm guarantee via Ville's inequality, and a two-channel controller whose frozen channel preserves that guarantee while an adaptive channel reads the drift rate (Section 4.4). The same martingale *certifies* the replay correction—its growth rate bounds the selector's coverage gap (Lemma 4.3).

(4) **Experiments.** On a diagnostic bandit and standard cooperative JaxMARL benchmarks, compatibility *selection* starves exploration (confirming Theorem 4.2), whereas *detection* flags co-learner drift faster than a TD-error heuristic at matched false-alarm (Section 5).

2 BACKGROUND AND NOTATION

2.1 Reinforcement learning

A Markov Decision Process (MDP) is a tuple (S, A, P, R, γ) with state set S, action set A, transition kernel $P(\cdot \mid s, a) \in \Delta(S)$, reward distribution $R(\cdot \mid s, a)$, and discount $\gamma \in [0, 1)$. A policy $\pi : S \to \Delta(A)$ induces a return $\sum_{t\geq 0} \gamma^t R_t$, and the action-value function $Q_\pi(s, a) = \mathbb{E}_\pi[\sum_{t\geq 0} \gamma^t R_t \mid S_0 = s, A_0 = a]$. The optimal value $Q^* = \max_\pi Q_\pi$ is the unique fixed point of the optimal Bellman operator

$$(\mathcal{B}^* Q)(s, a) = \mathbb{E}_{s'\sim P(\cdot|s,a)}\Big[R(s, a, s') + \gamma \max_{a'} Q(s', a')\Big], \quad (1)$$

which is a γ-contraction for $\|\cdot\|_\infty$ (Appendix A). Note that the right-hand side uses Q, not Q^*: $\mathcal{B}^*$ maps an arbitrary Q to a new function, and iterating it converges to Q^* by Banach's theorem.

Deep Q-learning. DQN [8] parameterises Q_ϕ by a neural network, stores transitions in a replay buffer $\mathcal{D}$, and minimises, over uniformly sampled batches,

$$\mathcal{L}(\phi) = \mathbb{E}_{(s,a,r,s')\sim U(\mathcal{D})}\Big[\big(Q_\phi(s, a) - y\big)^2\Big], \quad y = r + \gamma \max_{a'} Q_{\bar\phi}(s', a'), \quad (2)$$

with $\bar\phi$ a periodically updated target network. A stochastic gradient step on a batch of size B reads

$$\phi \leftarrow \phi - \alpha \nabla_\phi \frac{1}{B} \sum_{i=1}^{B} \big(y_i - Q_\phi(s_i, a_i)\big)^2. \quad (3)$$

Exploration typically uses an ϵ-greedy policy: act greedily w.p. $1-\epsilon$ and uniformly at random w.p. ϵ.

2.2 Multi-agent reinforcement learning

A cooperative *stochastic game* is a tuple $(N, S, (\mathbb{U}_a)_{a=1}^N, \mathcal{T}, (r_a)_{a=1}^N, \gamma)$: N agents share the state $s \in S$, agent a picks $u_a \in \mathbb{U}_a$, the joint action $\mathbf{u} = (u_a, \mathbf{u}_{-a})$ drives the transition $\mathcal{T}(\cdot \mid s, \mathbf{u})$ and the reward $r_a(s, \mathbf{u})$. We write a for the agent under consideration and $-a$ for the others, so that $u_a \sim \pi_a(\cdot \mid s)$ and $\mathbf{u}_{-a} \sim \pi_{-a}(\cdot \mid s)$.

Independent Q-learning (IQL).. In IQL [16] each agent runs its own DQN, treating the others as part of the environment. Its loss is

$$\mathcal{L}_a(\phi_a) = \mathbb{E}_{(s,\mathbf{u},r,s')\sim U(\mathcal{D})}\Big[\big(Q_{\phi_a}(s, u_a) - y_a\big)^2\Big], \quad (4)$$
$$y_a = r_a + \gamma \max_{u'_a} Q_{\bar\phi_a}(s', u'_a).$$

IQL is simple and, despite its non-stationarity, underlies competitive value-decomposition methods such as VDN [14] and QMIX [10]; the mechanisms we develop apply to those methods unchanged, since each agent still maintains an individual Q-function. We adopt the fully observable, simultaneous-action setting for clarity; the partially observable case is discussed in Section 7.

3 THE REPLAY MISMATCH IN MARL

3.1 The perceived MDP and the best-response value

Fix agent a and freeze the others at a joint policy π_{-a}. From a's viewpoint the game collapses to an MDP $\mathcal{M}_a(\pi_{-a})$ with *perceived* transition and reward kernels

$$P^{\pi_{-a}}(s' \mid s, u_a) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi_{-a}(\cdot|s)}\big[\mathcal{T}(s' \mid s, u_a, \mathbf{u}_{-a})\big], \quad (5)$$
$$\bar r^{\pi_{-a}}(s, u_a) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi_{-a}(\cdot|s)}\big[r_a(s, u_a, \mathbf{u}_{-a})\big]. \quad (6)$$

The value that IQL implicitly chases is the optimal value of this perceived MDP, i.e. the *best response* of agent a to π_{-a}, defined as the fixed point of the best-response Bellman operator

$$(\mathcal{B}^{\mathrm{BR}}_{\pi_{-a}} Q)(s, u_a) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi_{-a}(\cdot|s),\, s'\sim\mathcal{T}}\Big[r_a(s, \mathbf{u}) + \gamma \max_{u'_a} Q(s', u'_a)\Big]. \quad (7)$$

We denote its fixed point $Q^{\mathrm{BR}}_a(s, u_a; \pi_{-a})$. For *fixed* π_{-a}, $\mathcal{B}^{\mathrm{BR}}_{\pi_{-a}}$ is a γ-contraction (Appendix B), so $Q^{\mathrm{BR}}_a(\cdot; \pi_{-a})$ is well defined.

This is a *best-response* value, *not* a joint-optimal or equilibrium value: it is optimal for *a only against the fixed* π_{-a}. During training the others keep learning, so the relevant target is $Q^{\mathrm{BR}}_a(\cdot; \pi^{\mathrm{cur}}_{-a})$ with π^{cur}_{-a} *moving*. Off-policy MARL thus rests on a *quasi-stationary* approximation: at each update we treat the current others' policies as fixed. Perceived non-stationarity is precisely the drift of this target, and it is what makes a fixed buffer inconsistent.

3.2 What exactly is mismatched

The buffer stores transitions $\tau_i = (s_i, u_{i,a}, \mathbf{u}_{i,-a}, r_i, s'_i)$ in which the others' actions were drawn from their *past* policy, $\mathbf{u}_{i,-a} \sim \pi^{\mathrm{past}}_{i,-a}(\cdot \mid s_i)$. The loss we would *like* to minimise draws them from the *current* policy:

$$\mathcal{L}^{\mathrm{cur}}_a(\phi_a) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\mathrm{cur}}_{-a}(\cdot|s)}\Big[\big(Q_{\phi_a}(s, u_a) - y_a\big)^2\Big], \quad (8)$$
$$\mathcal{L}^{\mathrm{past}}_a(\phi_a) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\mathrm{past}}_{-a}(\cdot|s)}\Big[\big(Q_{\phi_a}(s, u_a) - y_a\big)^2\Big], \quad (9)$$

where s and u_a are drawn from the buffer in both lines, while the target y_a, the reward r and the next state s' are *downstream* of $\mathbf{u}_{-a}$

(since $r = r_a(s, u_a, \mathbf{u}_{-a})$ and $s' \sim \mathcal{T}(\cdot \mid s, u_a, \mathbf{u}_{-a})$). Equations (8) and (9) thus differ only through the conditional law of $\mathbf{u}_{-a}$ given s; the environment's conditional law of (r, s') given $(s, u_a, \mathbf{u}_{-a})$ is policy-independent, which is exactly why a single change of measure on $\mathbf{u}_{-a}$ (below) suffices. In particular, the *state* distribution is, in both cases, that of the buffer: reweighting the others' actions does *not* turn the past state-visitation $d^{\text{past}}(s)$ into the current one $d^{\text{cur}}(s)$. Any action-level correction therefore fixes *which joint actions* are credited, not *which states* are visited—a distinction that matters under function approximation and that we make explicit rather than hide.

3.3 Importance reweighting, done correctly

Going from (9) to (8) is a change of measure. For any integrable f,

$$\mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\text{cur}}_{-a}(\cdot|s)}[f] = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\text{past}}_{-a}(\cdot|s)}[\rho\, f], \qquad \rho = \frac{\pi^{\text{cur}}_{-a}(\mathbf{u}_{-a} \mid s)}{\pi^{\text{past}}_{-a}(\mathbf{u}_{-a} \mid s)}, \tag{10}$$

provided the support condition $\operatorname{supp}\pi^{\text{cur}}_{-a}(\cdot \mid s) \subseteq \operatorname{supp}\pi^{\text{past}}_{-a}(\cdot \mid s)$ holds. With unmasked ϵ-greedy exploration every action keeps positive probability, so ρ is finite; but when actions are *masked* (as in our Water-Bomber environment) the past support can miss actions that are likely now, and ρ diverges. This is not a numerical detail—it is where the method breaks.

Direct loss correction is unbiased but unstable. Multiplying each squared TD error by ρ_i yields an unbiased estimate of (8). Unfortunately ρ is a ratio of products of small probabilities over several agents; it is heavy-tailed, and a single large ρ_i produces a gradient spike. This is the empirically observed instability [3] that we revisit in Section 5.

Prioritisation is self-normalised, not unbiased. Instead of multiplying the loss, one can *sample* transition i with probability $P(i) = \rho_i / \sum_k \rho_k$ and drop the explicit factor. The resulting estimator of $\mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\text{cur}}_{-a}}[f]$ is the *self-normalised* importance estimator

$$\widehat{\mathbb{E}}_{\text{SNIS}} = \frac{\sum_i \rho_i f_i}{\sum_k \rho_k} \;\neq\; \frac{1}{N}\sum_i \rho_i f_i = \widehat{\mathbb{E}}_{\text{IS}}. \tag{11}$$

The two coincide asymptotically when $\mathbb{E}[\rho] = 1$, but the self-normalised one is *biased at finite batch size.* Crucially, the recommendation *not* to divide back by $P(i)$ (unlike prioritised experience replay [13]) is *not* a variance trick that preserves the objective: it is a *deliberate change of the optimised objective* towards the current-policy expectation. Stating this honestly removes the apparent contradiction between "we correct the expectation" and "we change the distribution without correction": we do the latter, on purpose.

Why the explicit ratio is not enough. Computing ρ_i requires (i) storing or recomputing $\pi^{\text{past}}_{i,-a}(\mathbf{u}_{i,-a} \mid s_i)$ and (ii) evaluating $\pi^{\text{cur}}_{-a}(\mathbf{u}_{i,-a} \mid s_i)$. This is feasible only for *queryable, discrete* policies with shared support. It fails for partially observable agents (whose policies condition on histories), continuous actions (densities, not masses), parametric or noisy exploration, and black-box or human teammates. These are exactly the realistic cases. We now replace the ratio by a quantity that survives all of them.

4 CONFORMAL REPLAY

The common object of this section is a *non-conformity score* that grades how incompatible a stored transition is with the other agents' current behaviour, and its calibration into distribution-free p-values. The score then drives both a *selection* rule (Section 4.3) and a *detection* rule (Section 4.4).

4.1 A non-conformity score for stored transitions

We seek a score $A(\tau)$ that is *large* when transition τ is stale, i.e. when its other-agent behaviour is unlikely under the present. We give three, in decreasing order of the access to the other agents they require.

(A) Policy-based scores. When the policies are queryable and the others explore with discrete, full-support actions, the *exact* score is the log importance ratio

$$A(\tau_i) = -\log\rho_i = -\log\pi^{\text{cur}}_{-a}(\mathbf{u}_{i,-a} \mid s_i) + \log\pi^{\text{past}}_{i,-a}(\mathbf{u}_{i,-a} \mid s_i)\,: \tag{12}$$

resampling transitions with weight ρ_i realises the self-normalised estimator (11), i.e. makes the replayed $\mathbf{u}_{-a}$ follow π^{cur}_{-a}, and recovers the importance-weighting of Foerster et al. [3]. It requires the *past* likelihoods $\pi^{\text{past}}_{i,-a}(\mathbf{u}_{i,-a} \mid s_i)$ to be stored with each transition.

When only the *current* policies can be queried, a cheaper, ratio-free surrogate is

$$A(\tau_i) = -\log\pi^{\text{cur}}_{-a}(\mathbf{u}_{i,-a} \mid s_i), \tag{13}$$

which scores high any stored joint action the others would now rarely take, needs no past likelihood, and has no support/blow-up problem. It is, however, a *coarser heuristic*: unlike $-\log\rho$ it does not target π^{cur}_{-a} exactly, and it over-penalises actions that are intrinsically rare yet still current (where $\pi^{\text{cur}}_{-a} \approx \pi^{\text{past}}_{-a}$ are both small, so $\rho \approx 1$). It is the fallback—together with the model-based score below—for the regimes (partial observability, continuous actions, black-box teammates) where ρ is unavailable.

(B) Model-based score. When the policies cannot be queried, learn a model $\hat{h}^{\text{cur}}(s', r \mid s, u_a)$ of the *currently perceived* dynamics from a sliding window of fresh transitions, and score a stored transition by its surprise,

$$A(\tau_i) = -\log\hat{h}^{\text{cur}}(s'_i, r_i \mid s_i, u_{i,a}), \tag{14}$$

or, for a mean predictor $\hat{m}^{\text{cur}}$ with residual covariance $\hat{\Sigma}$, by the Mahalanobis residual $A(\tau_i) = \|s'_i - \hat{m}^{\text{cur}}(s_i, u_{i,a})\|^2_{\hat{\Sigma}^{-1}}$. Because the others' policies enter the perceived kernel $P^{\pi^{\text{cur}}_{-a}}$, a change in π^{cur}_{-a} shows up as elevated prediction error: stale transitions are exactly the out-of-distribution ones. We deliberately do *not* assume the error has zero mean—a learned model is imperfect—and calibrate its nominal distribution instead, as follows.

4.2 From scores to conformal p-values

Maintain a calibration set $C = \{\tau^{\text{cal}}_j\}_{j=1}^n$ of *fresh* transitions, generated under the current policies and held out from fitting the score. Let $A^{\text{cal}}_j = A(\tau^{\text{cal}}_j)$. For a buffer transition τ_i with score $A_i = A(\tau_i)$, define the (inductive) conformal p-value

$$p_i = \frac{1 + \left|\{\, j :\; A^{\text{cal}}_j \geq A_i \,\}\right|}{n+1} \in (0, 1]. \tag{15}$$

Algorithm 1 Conformal Likely-Transition Selection (per agent a)

Require: buffer $\mathcal{D}$, fresh calibration set C, large-batch size L, mini-batch size B, score $A(\cdot)$

while training **do**
  refresh C with recent on-policy transitions; compute $\{A_j^{\text{cal}}\}$
  sample a large batch $\{\tau_i\}_{i=1}^{L}$ from $\mathcal{D}$ (uniformly or with PER)
  compute scores $A_i = A(\tau_i)$ and p-values p_i via (15)
  set weights $w_i \propto p_i$ (or $w_i \propto \mathbf{1}[p_i \geq \alpha]$); subsample B indices $\sim w$
  compute up-to-date targets y_i and take an SGD step (3) on the mini-batch
end while

A small p_i means τ_i is more non-conforming than almost all fresh transitions. Under the null hypothesis that τ_i is exchangeable with the calibration sample (its other-agent behaviour conforms to the present), p_i is (super-)uniform on $(0, 1]$ [21]. The construction is *distribution-free*: it inherits no Gaussian or zero-mean assumption from the score. When precision matters, a Mondrian / state-conditional variant restricts the calibration set to transitions whose (s, u_a) falls in the same region as τ_i, which guards against the model being merely less accurate in some part of the space.

4.3 Use 1: Conformal transition selection

We turn the p-values into a replay distribution. Following the large-batch recipe of LaBER [5], we draw a large batch, score it, and *subsample* a training mini-batch with a probability that *increases with conformity*, e.g. proportional to p_i, or a hard conformal filter $\mathbf{1}[p_i \geq \alpha]$ at level α (Algorithm 1). Intuitively we keep transitions that the present could plausibly have produced.

Relation to the importance ratio. With the exact score (12), conformal selection recovers a *rank-stabilised* form of the importance prioritisation of Foerster et al. [3]: a high ρ gives a low A, hence a high conformity p-value and a high selection weight. The difference is that the conformal calibration uses only the *rank* of A_i among fresh transitions, so the scale of ρ—and its heavy right tail (Section 3.3)—no longer destabilises training. The ratio-free score (13) and the model-based score extend the same rule to the cases where ρ cannot be computed.

What is and is not corrected. As in (11), this selection *deliberately* reshapes the training distribution towards the current perceived MDP; it is a calibrated heuristic reweighting, not an unbiased estimator, and—like every action-level scheme—it does not correct the state-visitation $d^{\text{past}}(s)$. We do not over-claim beyond this.

4.4 Use 2: Detecting drift of the perceived environment

Selection decides *which* transitions to trust; we also want to know *when* the buffer as a whole has gone stale, so as to refresh the calibration set, down-weight or partially flush the buffer, or widen exploration. This is a sequential change-detection problem on the stream of incoming p-values, for which conformal *test martingales* are the natural tool [9, 20, 22].

Algorithm 2 Conformal Drift Detection (per agent a)

Require: calibration set C, level δ, mixing over ε

$M \leftarrow 1$
for each newly observed transition τ_t **do**
  compute $A_t = A(\tau_t)$ and p_t via (15) against C
  $M \leftarrow M \cdot \int_0^1 \varepsilon\, p_t^{\varepsilon-1}\, \mathrm{d}\varepsilon$
  if $M \geq 1/\delta$ **then**
    declare drift: refresh C, down-weight/flush $\mathcal{D}$, reset $M \leftarrow 1$
  end if
end for

For a fixed $\varepsilon \in (0, 1)$ and p-values $p_1, p_2, \ldots$, define the power martingale, and its mixture,

$$M_t^{\varepsilon} = \prod_{j=1}^{t} \varepsilon\, p_j^{\varepsilon-1}, \qquad M_t = \int_0^1 \prod_{j=1}^{t} \varepsilon\, p_j^{\varepsilon-1}\, \mathrm{d}\varepsilon. \tag{16}$$

If, under the null H_0 (no drift), the p_j are conditionally super-uniform given the past, then $(M_t)_{t\geq 0}$ is a non-negative martingale with $M_0 = 1$ and $\mathbb{E}_{H_0}[M_t] = 1$ (Appendix D). Ville's inequality then bounds the *anytime* false-alarm probability,

$$\mathbb{P}_{H_0}\left(\sup_{t\geq 1} M_t \geq \tfrac{1}{\delta}\right) \leq \delta, \tag{17}$$

so raising an alarm when $M_t \geq 1/\delta$ controls the probability of *ever* falsely declaring drift by δ, with no need to fix a horizon (Algorithm 2).

Honest statement of guarantees. The bound (17) is *exact* only when the p_t are conditionally super-uniform under H_0. In RL this requires care: transitions are temporally dependent and the agent's own policy shapes the state distribution. We distinguish three regimes. *(i)* With an exact perceived model, or with inductive conformal calibration that preserves exchangeability [20], conditional validity holds and (17) is rigorous. *(ii)* With a held-out in-distribution calibration set we obtain marginal validity, which degrades if deployment visits regions under-represented in calibration. *(iii)* Otherwise the threshold is tuned on nominal trajectories for an acceptable empirical false-alarm rate. We report which regime we are in rather than claiming the strongest one by default.

4.5 Use 3: Certifying a replayed batch under a known shift

The p-values of (15) are valid only while the calibration set is exchangeable with the scored transitions. When the others' policies have *measurably* shifted—after the detector of Section 4.4 fires, or whenever the ratio ρ is itself available—we can *certify* validity rather than assume it, with two classical corrections that transfer directly to the replay stream. *(i) Covariate-shift weighting.* If the shift is captured by a likelihood ratio $w = \rho = \pi_{-a}^{\text{cur}}/\pi_{-a}^{\text{past}}$, the *weighted* conformal p-value [17]

$$p_i = \sum_j \tilde{w}_j\, \mathbf{1}\left[A_j^{\text{cal}} \geq A_i\right] + \tilde{w}_{\text{test}}, \qquad \tilde{w}_j = \frac{w_j}{\sum_k w_k + w_{\text{test}}}, \tag{18}$$

restores super-uniformity with respect to the *shifted* (current) law. *(ii) Time-decay (NexCP).* When the drift is gradual and the ratio is

unavailable, geometric weights $w_i = \rho_{\text{decay}}^{n-i}$ that down-weight older calibration points give the non-exchangeable conformal p-value of Barber et al. [2], whose coverage obeys the one-sided guarantee miscoverage $\le \alpha + \sum_i \tilde{w}_i\, d_{\text{TV},i}$. Both keep the selection and detection machinery *certified* under quantified drift—the constructive counterpart of the coverage gap bounded in Section 4.6—and we validate them in Section 5.

4.6 Theoretical analysis: when selection starves, and how detection certifies it

We make two things precise: *(i)* why biasing replay towards currently-compatible transitions can move the learned fixed point *away* from the best response, and *(ii)* in what sense the test martingale of Section 4.4 *certifies* the selection rule of Section 4.3.

Setup. Fix π_{-a}^{cur}, so the perceived MDP $\mathcal{M}_a(\pi_{-a}^{\text{cur}})$ is stationary (Appendix B). Use linear features $Q_\theta(s,u_a) = \phi(s,u_a)^\top\theta$ with Φ of full column rank—taken *tabular or $\mathcal{S}_w$-local* for the cleanest statements below, since a general Φ couples in off-support next-state features through the bootstrap. A buffer measure μ over (s,u_a) is reweighted by conformal weights $w_i = g(p_i) \ge 0$ (soft $w_i \propto p_i$, or the hard filter $w_i = \mathbf{1}[p_i \ge \alpha]$); write the effective (s,u_a)-marginal $d_w \propto w\cdot\mu$, $D_w = \mathrm{diag}(d_w)$, and the D_w-weighted projection $\Pi_{D_w} = \Phi(\Phi^\top D_w\Phi)^{-1}\Phi^\top D_w$. We write the reweighted operator with the *current* kernel $P^{\pi_{-a}^{\text{cur}}}$ and reward $\bar{r}^{\pi_{-a}^{\text{cur}}}$; this presupposes the exact-ratio correction (Theorem 4.2(iii)). Under ratio-free scores (13) or hard filtering the replayed (s',r) instead follow π_{-a}^{past}, so the effective kernel is the *past* one—a second bias compounding the support effect below.

Proposition 4.1 (Reweighted fixed point). *Let $A_w := \Phi^\top D_w(I-\gamma P^{\pi_{-a}^{\text{cur}}})\Phi$ and $b_w = \Phi^\top D_w\,\bar{r}^{\pi_{-a}^{\text{cur}}}$. If A_w is positive definite, conformal-reweighted semi-gradient TD converges to the unique fixed point $\theta_w = A_w^{-1}b_w$, solving $\Phi\theta_w = \Pi_{D_w}\mathcal{B}^{\text{BR}}_{\pi_{-a}^{\text{cur}}}(\Phi\theta_w)$; if A_w is merely non-singular the fixed point exists and is unique but the iteration need not converge to it. A* sufficient *(not necessary [4]) condition for $\Pi_{D_w}\mathcal{B}^{\text{BR}}_{\pi_{-a}^{\text{cur}}}$ to be a γ-contraction in $\|\cdot\|_{D_w}$ is that d_w be stationary for $P^{\pi_{-a}^{\text{cur}}}$; conformal reweighting deliberately violates it, so contraction—and hence boundedness of the off-distribution fixed point [4, 18]—is not guaranteed, and restoring it requires an emphatic reweighting [15].*

Theorem 4.2 (Support starvation). *Let $\mathcal{S}_w = \mathrm{supp}(d_w) \subseteq \mathrm{supp}(\mu) = \mathcal{S}$, with Φ tabular or $\mathcal{S}_w$-local.* (i) *θ_w depends on $\mathcal{M}_a(\pi_{-a}^{\text{cur}})$ only through the rows of A_w, b_w indexed by $\mathcal{S}_w$ (entries off $\mathcal{S}_w$ vanish).* (ii) *Hence $\Phi\theta_w$ solves the D_w-weighted projected Bellman equation supported on $\mathcal{S}_w$—the best response of the sub-MDP retaining $\mathcal{S}_w$; if an optimal best-response trajectory visits some $(s,u_a)\notin\mathcal{S}_w$, the greedy policy of θ_w can be arbitrarily suboptimal in $\mathcal{M}_a(\pi_{-a}^{\text{cur}})$.* (iii) *The hard filter $w_i = \mathbf{1}[p_i \ge \alpha]$ deletes from $\mathcal{S}_w$ every stored (s,u_a) whose $\mathbf{u}_{i,-a}$ is improbable under π_{-a}^{cur}; when the reward-bearing optimal joint action is currently low-probability (exploration, or escaping a poor consensus) it is deleted and (ii) applies. Conversely, the exact score $A = -\log\rho$ with soft weights $w_i \propto \rho_i$ (no truncation) leaves the (s,u_a)-marginal unchanged—$d_w = \mu$, since $\mathbb{E}_{\pi_{-a}^{\text{past}}}[\rho \mid s,u_a] = 1$, so $\mathcal{S}_w = \mathcal{S}$—and instead replaces the past perceived kernel by the current one inside the Bellman operator (the importance correction of Foerster et al. [3]), so θ_w targets $Q_a^{\text{BR}}(\cdot;\pi_{-a}^{\text{cur}})$.*

Proof: Appendix E. The theorem delimits the *safe regime* of conformal selection—soft, exact-ratio reweighting, which both preserves support and corrects the kernel, recovering 3. Outside it—ratio-free scores (13) or hard α-filtering—selection both starves support *and* leaves the kernel biased towards π_{-a}^{past}.

Remark (the Simultaneous-Attack collapse). In Simultaneous-Attack (single state; Section 5) the all-attack joint action is the only reward-bearing event. Once the iterate drifts towards all-abstain, π_{-a}^{cur} renders all-attack improbable and the hard filter zeroes its weight, so the rare exploratory transitions leave $\mathcal{S}_w$; by Theorem 4.2(iii) learning then locks onto the all-abstain consensus—predicting precisely the regime in which compatibility selection underperforms uniform replay. (With a single state the trajectory argument of (ii) degenerates; the operative mechanism is the weight-zeroing of the rewarding arm.)

Lemma 4.3 (Detection certifies selection). *Run the martingale of Section 4.4 on the* same *score A as the selector, with* fixed *(data-independent) weights $\tilde{w}_i$—as required by Barber et al. [2]; the data-dependent rule $w_i \propto p_i$ of Algorithm 1 is thus certified only in its frozen-weight form. Model the windowed p-values as i.i.d. with density q under drift (uniform under H_0)—an idealisation stronger than the conditional super-uniformity used elsewhere. Then the beyond-exchangeability coverage gap of the selector [2] obeys*

$$\Delta_{\text{cov}} \;\le\; \sum_i \tilde{w}_i\, d_{\text{TV}}\big(\mathbb{P}_A^{\text{cur}}, \mathbb{P}_A^{\text{past}}\big) \;\le\; \Big(\textstyle\sum_i \tilde{w}_i\Big)\sqrt{\tfrac{1}{2}\,\mathrm{KL}(q\,\|\,\mathrm{Unif})},$$

and the detector estimates the right-hand side: $\frac{1}{t}\log M_t \xrightarrow{a.s.} \mathbb{E}_q[\log g] = \mathrm{KL}(q\,\|\,\mathrm{Unif}) - \mathrm{KL}(q\,\|\,g) \le \mathrm{KL}(q\,\|\,\mathrm{Unif})$, with equality for the log-optimal bet $g = q$.

Proof: Appendix F. A *log-optimal* detector ($g = q$) makes $\frac{1}{t}\log M_t \to \mathrm{KL}(q\,\|\,\mathrm{Unif})$, an asymptotically valid *upper* bound on the selector's miscoverage. The deployed power mixture (16) is not log-optimal, so its growth rate *under*-estimates $\mathrm{KL}(q\,\|\,\mathrm{Unif})$ by the regret $\mathrm{KL}(q\,\|\,g)$: it certifies selection only *qualitatively* (faster growth $\Rightarrow$ larger drift $\Rightarrow$ wider potential gap). The bound is numerically tight only when the bet is log-optimal *and* $A = -\log\rho$ (a sufficient statistic of $\pi_{-a}^{\text{past}} \to \pi_{-a}^{\text{cur}}$, so that score drift equals perceived-MDP drift); and the growth-rate estimate is asymptotic, whereas the alarm (17) is the anytime-valid object. In this precise sense, *detection certifies selection.*

5 EXPERIMENTS

5.1 Environments

All environments are implemented in JAX, so every method runs inside a single `lax.scan` training loop vectorised across seeds. A custom one-step bandit isolates the mechanism; three cooperative JaxMARL [11] benchmarks place it in standard settings. The drift-*detection* experiments use a fixed partner-swap intervention and a calibration set that both assume a state-independent action set, so they run on the unmasked envs (Simultaneous-Attack, MPE, Overcooked); SMAX—whose legal actions are state-dependent—instead exercises the masked-action training target (Section 3.3) and the selection ablation.

Simultaneous-Attack. A one-step, single-state bandit with N agents and two actions, *attack* or *abstain*. Attacking together succeeds; attacking without full participation is "suicidal". With $K = \sum_i a_i$ attackers we use the reward

$$r(\mathbf{a}) = \begin{cases} 1 & \text{if } K = N, \\ -r_{\text{death}}\, K & \text{if } 0 < K < N, \\ 0 & \text{if } K = 0, \end{cases} \qquad r_{\text{death}} = \frac{1}{N\,(2^{N-1}-1)}. \tag{19}$$

This constant (corrected from the earlier $-1/(1-2^{-(N-1)})$, which had the wrong sign and scale) makes *abstention-vs-coordination* the whole difficulty: a uniformly random joint policy gets expected return 0, while the all-attack optimum gets 1 (proof in Appendix C). Coordination is exponentially rare under random play, so the buffer fills with "failed" transitions whose value depends entirely on whether teammates would still attack.

MPE `simple_spread`. The standard cooperative-navigation benchmark [6] (agents spread to cover landmarks): a multi-state, partially observable setting in which the perceived dynamics genuinely shift when a teammate's policy changes—our drift-detection testbed.

SMAX.. The StarCraft-style micromanagement challenge [12] reimplemented in JAX, with *state-dependent action masks*—the regime in which ρ loses support (Section 3.3). It carries the masked-action double-Q target and the selection ablation.

Overcooked. A fully-cooperative coordination benchmark in which a scripted *partner swap* at a known step injects a controlled drift event.

5.2 Protocol

The backbone is an IQL agent in JAX on top of JaxMARL—a recurrent Q-network with a Flashbax replay buffer, trained with the PureJaxRL `scan+vmap` recipe so all seeds run in one compiled call; ϵ is annealed from 0.99 to 0.05. Replay sampling is a swappable module—UNIFORM, PER, LaBER, exact-ratio ($A = -\log\rho$), and the conformal policy/model scores—over a shared loop, so ablations differ only in the sampler. Conformal scores are calibrated on a fresh on-policy hold-out; the drift martingale uses the ε-grid mixture at level $\delta = 0.05$. We report *average optimality* (normalised return) for selection and *detection delay at matched false-alarm* for detection, over 8 seeds (more on the cheap bandit), with inter-seed spread shaded.

5.3 Findings

Selection by compatibility starves exploration—the negative that proves the theory. On Simultaneous-Attack ($N = 3$), UNIFORM replay reaches the optimum on *every* seed at buffer sizes {2k, 50k, 200k} (final optimality 1.00), whereas compatibility selection *collapses* a fraction of seeds: the ratio-free conformal score falls to 0.66 ± 0.49 and even the exact ratio $A = -\log\rho$ to 0.57 ± 0.51, both bimodal (12 seeds; Figure 2). The conformal machinery is not at fault—the p-values are super-uniform under H_0 (Kolmogorov–Smirnov statistic ≈ 0.006); the cause is precisely Theorem 4.2(iii), as deleting the rare exploratory all-attack transitions pushes the reward-bearing action out of $\mathcal{S}_w$. We confirm the theorem's *iff* structure directly (16 seeds, support floor 0.3; Figure 3): selection recovers the optimum (1.00, 0/16 collapse) *only* when the exact-ratio score *and* a support floor both hold; dropping either returns to 0.75 (4/16)—isolating the theorem's two conditions and ruling out a trivial "a floor fixes everything" reading. Across samplers the per-seed correlation between surviving all-attack mass and final return is +0.998. Selection *by filtering* is thus not the contribution; its certified, support-preserving forms are (below).

Detection is the robust contribution. The two-channel controller pairs a *frozen* calibration channel—whose Ville alarm preserves the anytime false-alarm guarantee—with an adaptive channel that reads the log-wealth slope $\mathrm{d}\log M_t/\mathrm{d}t$ ($\approx$ the KL drift rate) to drive the buffer half-life. The frozen channel is anytime-valid in practice: over 20,000 null streams the empirical *ever*-cross rate is 0.040 at $\delta = 0.1$ and 0.019 at $\delta = 0.05$, at or below the Ville bound (17). On MPE `simple_spread`, after a controlled partner-policy swap, the conformal martingale detects 8/8 seeds at delay 21–29 (mean 25.5) at *zero* false alarms, using a *single threshold transferred* from the Simultaneous-Attack false-alarm lab. The TD-error heuristic AES [23], held to that *same* transferable threshold, fires in only 4/8 seeds; with a task-local reference it fires in 0/8, and only under full per-environment recalibration does it reach 6/8—then slowly (median delay 86) and still missing two seeds (Figure 4). Conformal detection thus pairs an anytime guarantee with a threshold that *transfers*, the asymmetry Lemma 4.3 anticipates. The Overcooked partner-swap corroborates this (7/8, median delay $\approx$120) at a smaller training budget (Figure 5).

Certified replay recovers what filtering loses. Where ρ is available, ρ-weighted conformal p-values [17] give the exact-ratio score its first finite-sample validity under the known co-action shift—the "missing middle" between the ratio-free heuristic and raw importance sampling—and time-decay NexCP [2] turns Foerster's heuristic discount of stale data into one whose one-sided coverage gap vanishes with the drift. On controlled score streams with a *known* shift the gains are exact: under a covariate shift the ordinary inductive p-value is badly miscalibrated ($\mathbb{P}(p \le 0.05) = 0.26$, a 5.2× inflation), whereas the ρ-weighted p-value (18) restores it to 0.04 (mean p: $0.24 \to 0.50$); for a gradually drifting stream, plain exchangeable conformal under-covers at 30.5% against the 80% nominal while time-decay NexCP weighting restores 85.8%, the one-sided bound holding throughout. Both are super-uniform with coverage tending to the nominal level as drift $\to 0$, matching Lemma 4.3.

Forgetting-based replay control hurts in this setting. The tempting way to *act* on the detector is to forget: when the frozen alarm fires, flush the stale buffer, and let the drift rate shorten the buffer half-life. We test whether this improves *return* recovery after a co-learner drift on MPE `simple_spread`. A single ego agent learns by IQL while its partners—coherent before a known step—switch to a non-cooperative scripted policy after it, a controlled swap that both lowers the team return and is strongly non-conforming. Policies A/B are pretrained once and shared across the 12 recovery/environment RNG runs (the scripted post-swap arm uses a fixed action); the reported SEM are therefore conditional on this fixed pretraining and do not measure end-to-end training variability. The detector behaves exactly as designed, firing on 12/12 seeds at a mean delay

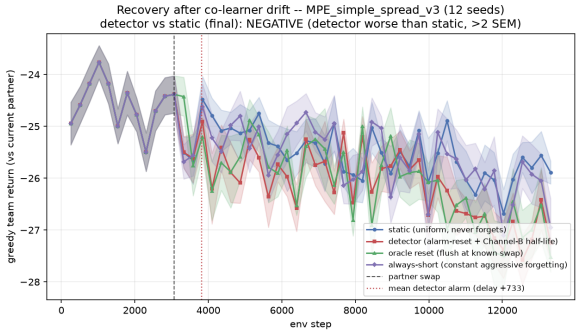


Figure 1: Return recovery after a co-learner drift (MPE `simple_spread`, 12 seeds). The detector fires reliably (mean delay +733 steps, dotted line). Each forgetting-based replay control—the two-channel controller (alarm-flush + half-life), an oracle flushing at the *known* swap, and a constant short half-life—has lower mean final return than a STATIC buffer that keeps all data. The pattern is consistent with loss of useful replay coverage, but Theorem 4.2 does not directly cover age-weighted nonlinear replay. Bands: ±1 inter-seed SEM.

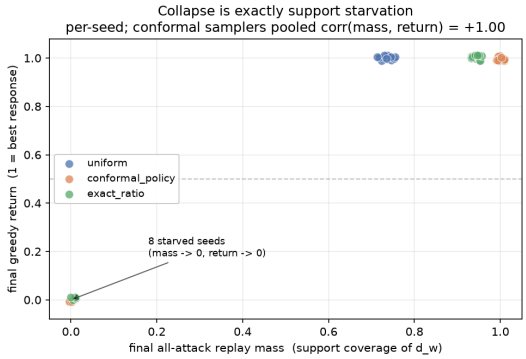


Figure 2: Support starvation (Simultaneous-Attack). UNIFORM replay reaches the optimum on every seed; compatibility selection collapses a fraction of seeds by deleting the rare exploratory transitions that carry the reward (Theorem 4.2(iii)). Shaded band: inter-seed spread.

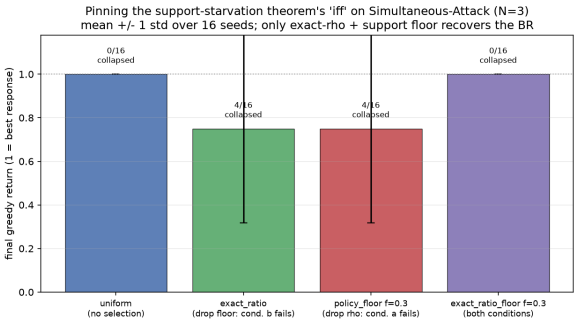


Figure 3: The theorem's two conditions, isolated (16 seeds). Final optimality is 1.0 only for UNIFORM and for exact-ratio reweighting *with* a support floor (both conditions met); dropping the exact ratio (policy floor) or the floor (exact ratio alone) returns to 0.75. Support coverage *and* the exact ratio are each necessary.

of 733 steps; but acting on it does *not* help. Against a STATIC buffer that keeps all data, the alarm-flush-plus-half-life controller is *worse*: the paired detector-minus-static differences are $\Delta_{\text{final}} = -1.28 \pm 0.31$ and $\Delta_{\text{AUC}} = -0.60 \pm 0.17$ (conditional mean ± SEM), both beyond 2 conditional SEM. The ranking is monotone in how much each arm forgets (STATIC > constant short half-life > oracle reset ≈ controller). An *oracle* that flushes at the *known* change point performs similarly to the detector, making detection delay an unlikely sole explanation (Figure 1). The pattern is consistent with an empirical temporal analogue of the support-loss mechanism in Theorem 4.2: flushing or aggressively discounting old data can reduce useful coverage. It is not a formal instance of the theorem, which assumes tabular or local linear features and zero-weight support deletion; moreover the partner swap changes the learner's perceived kernel even though the underlying environment dynamics are fixed. The tested detector-triggered forgetting policy therefore worsens post-drift return in this setting, but the experiment does not establish the benefit of a corrective reweighting arm.

6 RELATED WORK

The non-stationarity of the perceived environment is a long-standing theme in cooperative MARL [7]. Foerster et al. [3] stabilise experience replay by conditioning on a low-dimensional fingerprint of training time and by importance-weighting the loss; we revisit their ratio, correct its formulation, and remove its dependence on queryable discrete policies. Value-decomposition methods such as VDN [14] and QMIX [10], and the actor-critic MADDPG [6], address credit assignment but still inherit non-stationary individual learners; PPO-based on-policy methods [24] sidestep replay at a sample-efficiency cost. On the replay side, prioritised experience replay [13] and LaBER [5] reshape the sampling distribution by TD error or gradient norm; we borrow their large-batch machinery but prioritise by *policy conformity* rather than by error magnitude. Our detection component builds on conformal prediction [1, 21] and on conformal test martingales for change detection [20, 22], whose anytime validity follows from Ville's inequality [9, 19].

7 CONCLUSION AND FUTURE WORK

We recast the reuse of past experience in off-policy MARL as a question of *conformity to the present* and answered it with conformal prediction. A single non-conformity score, calibrated into distribution-free p-values, *detects* drift of the perceived environment through a conformal test martingale—with an anytime false-alarm guarantee and a threshold that transfers across environments—and *certifies* the replayed batch under a quantified shift via weighted and time-decayed p-values. The same score can also *select* compatible transitions, but here our message is deliberately cautionary: we prove, and confirm empirically, that outside the exact-ratio regime

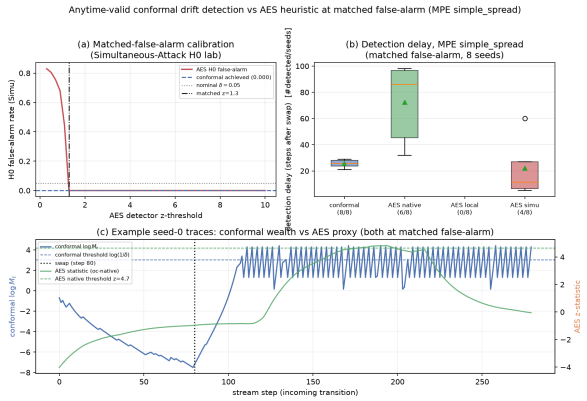


Figure 4: Drift detection on MPE `simple_spread` after a partner swap. With a single threshold transferred from the bandit false-alarm lab, the conformal martingale detects 8/8 seeds (mean delay 25.5) at zero false alarms; the AES TD-error heuristic, at the same threshold, misses seeds.

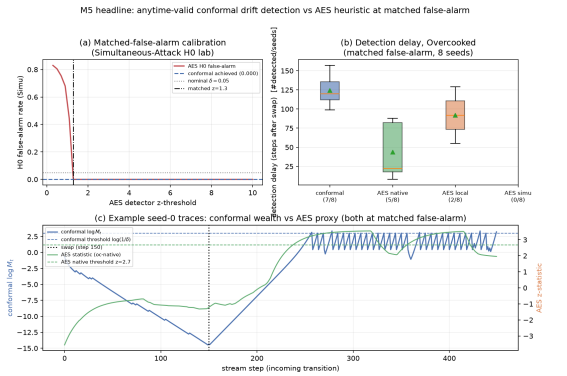


Figure 5: Two-channel controller on the Overcooked partner-swap: the frozen Ville-alarm channel (anytime false-alarm $\leq \delta$) versus AES at matched false-alarm, with the adaptive channel's drift-rate estimate. Corroborates Figure 4 at a smaller budget.

selection starves the support it needs and underperforms uniform replay. Detection and certification are thus the robust contributions; selection is safe only in the narrow regime where it coincides with the (often unavailable) importance ratio. The caution has an empirical temporal analogue: driving replay to *forget* on a detected drift—flushing the buffer or shortening its half-life—worsens post-drift return in our MPE study. This result does not formally extend Theorem 4.2 to age-weighted replay, nor does it test corrective reweighting on the recovery task. Scaling this recovery study to higher-coordination benchmarks such as Overcooked, where partner conventions matter more, is left to future work.

Several directions remain. The immediate next step is full-scale convergence on the masked / partially observable testbeds—SMAX and Overcooked—where the importance ratio loses support and our stored-state GRU detector already runs; these runs are GPU-bound and in progress. The state-visitation mismatch is untouched and could be addressed with state-conditional (Mondrian) calibration or corrections on $d(s)$; validity in the dependent RL stream deserves a dedicated analysis via e-processes [9]; and value-decomposition backbones such as VDN and QMIX inherit the same per-agent score unchanged.

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A CONVERGENCE OF Q-LEARNING, CORRECTED

The optimal Bellman operator (1) maps an *arbitrary* Q to a new function (it uses Q on the right-hand side, not Q^*). It is a γ-contraction in $\|\cdot\|_\infty$:

$$\begin{aligned}\big|(\mathcal{B}^*Q_1)(s,a) - (\mathcal{B}^*Q_2)(s,a)\big| &= \gamma\Big|\mathbb{E}_{s'}\big[\max_{a'} Q_1(s',a') - \max_{a'} Q_2(s',a')\big]\Big| \\ &\le \gamma\,\mathbb{E}_{s'}\big[\max_{a'} |Q_1(s',a') - Q_2(s',a')|\big] \\ &\le \gamma\,\|Q_1 - Q_2\|_\infty,\end{aligned}$$

using $|\max_x f(x) - \max_x g(x)| \le \max_x |f(x) - g(x)|$. Hence $\|\mathcal{B}^*Q_1 - \mathcal{B}^*Q_2\|_\infty \le \gamma\|Q_1 - Q_2\|_\infty$ with $\gamma < 1$; the factor γ is what makes it a contraction (a *non-expansion* would not suffice for Banach's theorem). The fixed point is Q^*.

B THE PERCEIVED BEST-RESPONSE OPERATOR

For fixed π_{-a}, the operator $\mathcal{B}^{\mathrm{BR}}_{\pi_{-a}}$ of (7) is a γ-contraction: the others' actions are averaged out into the perceived kernel $P^{\pi_{-a}}$ and reward $\bar{r}^{\pi_{-a}}$, reducing (7) to an ordinary optimal Bellman operator on $\mathcal{M}_a(\pi_{-a})$, to which Appendix A applies verbatim. Its fixed point $Q^{\mathrm{BR}}_a(\cdot;\pi_{-a})$ is the best response of a to π_{-a}. The MARL Bellman optimality equation

$$Q^{\mathrm{BR}}_a(s,u_a;\pi^{\mathrm{cur}}_{-a}) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\mathrm{cur}}_{-a}}\Big[r_a(s,\mathbf{u}) + \gamma \sum_{s'} \mathcal{T}(s' \mid s,\mathbf{u}) \max_{u'_a} Q^{\mathrm{BR}}_a(s',u'_a;\pi^{\mathrm{cur}}_{-a})\Big]$$

must be read as a best-response value *conditional on the current others' policies*, under the quasi-stationary approximation that they are held fixed during the update—not as a joint-optimal or equilibrium value.

C SIMULTANEOUS-ATTACK REWARD, CORRECTED

Let each agent attack independently with probability $\frac{1}{2}$ (uniform random policy), so $K = \sum_i a_i \sim \mathrm{Binomial}(N, \frac{1}{2})$. With the reward (19),

$$\mathbb{E}_U[r] = \underbrace{1 \cdot \mathbb{P}(K = N)}_{=2^{-N}} - r_{\mathrm{death}} \sum_{k=1}^{N-1} k\,\mathbb{P}(K = k).$$

Since $\sum_{k=0}^{N} k\,\mathbb{P}(K=k) = \mathbb{E}[K] = N/2$ and the $k = N$ term is $N \cdot 2^{-N}$,

$$\sum_{k=1}^{N-1} k\,\mathbb{P}(K=k) = \frac{N}{2} - \frac{N}{2^N} = \frac{N}{2}\left(1 - 2^{-(N-1)}\right) = \frac{N\,(2^{N-1}-1)}{2^N}.$$

Therefore $\mathbb{E}_U[r] = 2^{-N} - r_{\mathrm{death}} \cdot N(2^{N-1}-1)\,2^{-N}$, which vanishes exactly for

$$r_{\mathrm{death}} = \frac{1}{N\,(2^{N-1}-1)} > 0,$$

giving a genuine *penalty* ($-r_{\mathrm{death}}K < 0$) for partial attacks, $\mathbb{E}_U[r] = 0$ for random play, and $r = 1$ for the all-attack optimum. (For $N = 2$: $r_{\mathrm{death}} = \frac{1}{2}$, and $\mathbb{E}_U[r] = \frac{1}{4} - \frac{1}{2} \cdot \frac{1}{2} = 0$.)

D CONFORMAL MARTINGALE VALIDITY

Let $\mathcal{F}_t = \sigma(p_1, \ldots, p_t)$. If, under H_0, each p_t is conditionally super-uniform, $\mathbb{P}(p_t \le u \mid \mathcal{F}_{t-1}) \le u$ for all $u \in [0,1]$, then for $\varepsilon \in (0,1)$,

$$\mathbb{E}\big[\varepsilon p_t^{\varepsilon-1} \mid \mathcal{F}_{t-1}\big] \le \int_0^1 \varepsilon u^{\varepsilon-1}\,\mathrm{d}u = 1,$$

so $M_t^\varepsilon = \prod_{j\le t} \varepsilon p_j^{\varepsilon-1}$ is a non-negative supermartingale with $M_0 = 1$ (a martingale when the p_t are exactly uniform). Mixing over ε preserves this. Ville's inequality [19] for non-negative supermartingales gives $\mathbb{P}_{H_0}(\sup_t M_t \ge 1/\delta) \le \mathbb{E}[M_0]\,\delta = \delta$, which is (17) and justifies the anytime threshold $M_t \ge 1/\delta$.

E PROOF OF THEOREM 4.2 (SUPPORT STARVATION)

Take Φ tabular or $\mathcal{S}_w$-local, so Π_{D_w} does not couple in features of states outside $\mathcal{S}_w$; for a general Φ the bootstrap term $\gamma\Phi^\top D_w P^{\pi^{\mathrm{cur}}_{-a}}\Phi$ carries next-state features $\phi(l)$ with $l \notin \mathcal{S}_w$, and the clean "sub-MDP" reading of (ii) holds only when $\mathcal{S}_w$ is closed under $P^{\pi^{\mathrm{cur}}_{-a}}$. *(i)* Both $A_w = \Phi^\top D_w(I - \gamma P^{\pi^{\mathrm{cur}}_{-a}})\Phi$ and $b_w = \Phi^\top D_w \bar{r}^{\pi^{\mathrm{cur}}_{-a}}$ carry $D_w = \mathrm{diag}(d_w)$ as a left factor; any (s,u_a) with $d_w(s,u_a) = 0$ contributes a zero row to D_w and so does not enter A_w or b_w. Hence θ_w is determined by $\{P^{\pi^{\mathrm{cur}}_{-a}}, \bar{r}^{\pi^{\mathrm{cur}}_{-a}}\}$ only through the rows indexed by $\mathcal{S}_w$. *(ii)* On $\mathcal{S}_w$ the normal equation $A_w\theta_w = b_w$ is the D_w-weighted projected Bellman equation of the sub-MDP that retains only state-actions in $\mathcal{S}_w$, whose solution is that sub-MDP's best-response value. Two perceived MDPs that agree on $\mathcal{S}_w$ but differ on $\mathcal{S} \setminus \mathcal{S}_w$ yield identical A_w, b_w, hence identical θ_w, while their true Q^{BR}_a differ; choosing the off-support values so that an omitted action is optimal makes $\arg\max_{u_a} \phi(s,u_a)^\top\theta_w$ suboptimal, with a gap as large as the reward range permits. *(iii)* For the hard filter $d_w(s,u_a) \propto \mu(s,u_a)\,\mathbf{1}[p \ge \alpha]$, and p is small exactly when $\mathbf{u}_{i,-a}$ is improbable under π^{cur}_{-a} (small $\pi^{\mathrm{cur}}_{-a}(\mathbf{u}_{i,-a} \mid s)$, large score A); such pairs receive $d_w = 0$ and leave $\mathcal{S}_w$. For the converse, ρ is a function of $\mathbf{u}_{-a}$ (not of (s,u_a)) with $\mathbb{E}_{\pi^{\mathrm{past}}_{-a}}[\rho \mid s,u_a] = 1$, so $w_i \propto \rho_i$ leaves the (s,u_a)-marginal unchanged ($d_w = \mu$, $\mathcal{S}_w = \mathcal{S}$) while, within each (s,u_a), re-weighting the stored $(\mathbf{u}_{-a}, s', r)$ from π^{past}_{-a} to π^{cur}_{-a}—replacing the past perceived kernel by the current one, on which the ordinary contraction of Appendix B applies and θ_w targets $Q^{\mathrm{BR}}_a(\cdot;\pi^{\mathrm{cur}}_{-a})$.

F PROOF OF LEMMA 4.3 (DETECTION CERTIFIES SELECTION)

Barber et al. [2], extending weighted conformal prediction [17], bound the coverage gap with *fixed* weights at the *residual* (score) level: coverage $\ge 1 - \alpha - \sum_i \tilde{w}_i\, d_{\mathrm{TV}}\big(R(Z), R(Z^{(i)})\big)$, where R is the score map and $Z^{(i)}$ swaps calibration point i with the test point. With $R = A$ and the windowed i.i.d. model each swap perturbs a

single coordinate, so $d_{\mathrm{TV}}(R(Z), R(Z^{(i)})) = d_{\mathrm{TV}}(\mathbb{P}_A^{\mathrm{cur}}, \mathbb{P}_A^{\mathrm{past}})$ and the bound collapses to the Lemma's first inequality. Let F_{past} be the (continuous, tie-free) past score CDF, so $p = 1 - F_{\mathrm{past}}(A)$ is a monotone, invertible probability integral transform: under H_0, $p \sim \mathrm{Unif}[0,1]$; under drift its law has density q. As KL and total variation are invariant under invertible maps, $\mathrm{KL}(q \,\|\, \mathrm{Unif}) = \mathrm{KL}(\mathbb{P}_A^{\mathrm{cur}} \,\|\, \mathbb{P}_A^{\mathrm{past}})$ and likewise for d_{TV}; Pinsker's inequality $d_{\mathrm{TV}} \le \sqrt{\frac{1}{2}\mathrm{KL}}$ then gives the second inequality. For the growth rate, $\frac{1}{t}\log M_t = \frac{1}{t}\sum_{j\le t}\log g(p_j) \to \mathbb{E}_q[\log g(p)]$ a.s. (SLLN), and $\mathbb{E}_q[\log g] = \mathrm{KL}(q \,\|\, \mathrm{Unif}) - \mathrm{KL}(q \,\|\, g) \le \mathrm{KL}(q \,\|\, \mathrm{Unif})$, with equality iff $g = q$. Hence only the log-optimal bet makes $\frac{1}{t}\log M_t \to \mathrm{KL}(q \,\|\, \mathrm{Unif})$ and turns the plug-in into an (asymptotic) *upper* bound on Δ_{cov}; the power mixture (16), equal to q only when q lies in the family $\{\varepsilon p^{\varepsilon-1}\}$, under-estimates it by the regret $\mathrm{KL}(q \,\|\, g)$ and certifies only qualitatively. The bound also requires fixed weights (the data-dependent selector $w_i \propto p_i$ falls outside the hypothesis), and equality of score drift with perceived-MDP drift requires A to be a sufficient statistic of $\pi_{-a}^{\mathrm{past}} \to \pi_{-a}^{\mathrm{cur}}$, i.e. $A = -\log \rho$. The estimate is asymptotic; the alarm (17) remains the anytime-valid object.